Joe Gasewicz

Senior Fullstack Python & Typescript Engineer
London, England, United Kingdom

https://github.com/joegasewicz

Professional Summary

Senior fullstack software engineer with extensive experience designing, building, and operating production-grade systems using Python, Golang, Dart, TypeScript, C & Java. Proven track record delivering data-intensive, cloud-hosted applications across finance, government, healthcare, and AI-driven environments. Also have substantial experience working as a front end developer, focusing on Typescript, Sass, HTML5, CSS, React & Angular.

Founder of josef.digital, open-source contributor, and framework author, with hands-on experience across AWS infrastructure, microservices, APIs, CI/CD, and DevOps. Experienced in building AI-powered systems, including developing agentic AI architectures and autonomous agents using Python and LangChain, integrating large language models such as Mistral, and deploying local and production inference pipelines using Ollama. Strong understanding of tool orchestration, memory systems, retrieval-augmented generation (RAG), and workflow automation.

Key Skills

- Python, Golang, Dart, TypeScript, C
- Django, Dash (Plotly), Flask, FastAPI, Tornado, ReactJS, SQLAlchemy
- AWS (Lambda, API Gateway, RDS, DynamoDB), Docker, PostgreSQL, GitHub CI, Terraform, Ansible, Caddy
- Microservices, RESTful APIs, Data Pipelines, Event-Driven Systems, RBAC & Auth, CI/CD, Observability
- Pytest, Jest, Cypress, Nightwatch.js, PySelenium, Gherkin, Cucumber, Postman,
- AI Product Engineering, Data Pipelines, Feature Engineering, Model-Serving APIs, Analytics Platforms, LLM & AI Agent Development

Open Source Contributions & Authored Projects

- JupyterHub
- Marshmallow
- Go-bindata-assetfs

- ForestMQ: Message Queue written
- Gomek: Go http framework.
- Bobtail: Lightweight WSGI Python HTTP framework
- Pytask-IO: Python async task queue
- Flask-file-upload: Easy file uploads for Flask
- flask-jwt-router: Auth library for Flask, including Google OAuth2.0
- react-google-oauth2.0: React frontend login with OAuth 2.0

Professional Experience

NHS

Senior Fullstack Developer (Python/Typescript/Angular)

April 2026 – Present

- Developed secure biometric authentication and authorisation capabilities for NHS Identity and Access Management (IAM) services.
- Built and enhanced staff onboarding systems using Python, FastAPI, Angular, and AWS, streamlining identity verification and access provisioning.
- Integrated the IAM AFCID service with the internal CIM platform to connect and manage users across NHS systems.
- Designed REST APIs and integration workflows between onboarding, identity, biometric, and workforce services.
- Collaborated with security, product, and operational teams to deliver reliable services aligned with NHS security and data-protection requirements.

Morgan Stanley

Senior Fullstack Developer (Python/Typescript/React)

June 2025 – January 2026

- Developed internal **data visualisation and analytics platforms** using Dash, supporting financial and operational decision-making.

- Built and maintained **high-performance backend services** in Python and Go for data-heavy workflows.

- Designed and implemented **FastAPI microservices**, exposing secure, well-documented APIs for internal consumers.

- Collaborated closely with quantitative, analytics, and engineering teams in a **high-governance, regulated environment**.

- Focused on reliability, performance, and long-term maintainability of production systems.
- Contributed to the main dashboard application built on AngularJS, ReactJS & FastAPI.

UK Health Security Agency
Senior Python / Django Software Engineer

December 2024 – April 2025

- Implemented middleware enforcing RBAC group context for a large-scale, public-facing Django application.

- Designed a flexible authorisation layer using a Fluent / Actions design pattern.
- Worked across backend services supporting data-intensive dashboards.

- Took ownership of QA, integration testing, and production readiness.

Post Office Ltd
Senior Python / Go Software Engineer
November 2022 – November 2024

- Designed and built **ETL pipelines** handling large volumes of data using Python and Go.

- Developed and deployed **serverless AWS architectures** (Lambda, API Gateway, RDS).

- Integrated PostgreSQL and DynamoDB to support **event-driven and transactional workloads**.

- Led performance testing initiatives using **Apache JMeter**, ensuring scalability and reliability under load.

Office for National Statistics
Senior Golang / Java / TypeScript Developer
January 2022 – October 2022

- Maintained and improved legacy front-end systems, strengthening authentication and security.

- Built a public-facing website backed by Golang services.

- Delivered **geospatial and data-driven features** using Leaflet.js and Mapbox.

- Contributed to the modernisation of the frontend stack and development practices.
- Migration from Java / Spring Boot to Go services.

SigTech
Senior Python & Go Developer
August 2021 – November 2021

- Designed a **distributed JupyterHub deployment** for a quantitative trading platform.

- Contributed to the **JupyterHub open-source project**.

- Optimised Tornado-based services for real-time, multi-user workloads.

- Integrated Kubernetes and Docker for scalable execution environments.

- Delivered E2E testing using PySelenium.

Mission Digital
Lead Full-stack Python & TypeScript Developer
September 2020 – August 2021

- Led the design and delivery of a **film file archiving platform**, from concept to production.

- Built RESTful APIs in Flask for managing high-resolution media and metadata.

- Developed a React/TypeScript frontend with a focus on usability and performance.

- Implemented authentication and **role-based access control**.

- Worked closely with non-technical stakeholders, iterating rapidly on requirements.

NHS
Senior Python / Django Developer
June 2020 – September 2020

- Contributed to the **NHS Track and Trace** application.

- Worked on authentication, access control, and user management with a focus on data privacy.

Bink
Senior Python & Go Developer
December 2019 – March 2020

- Developed middleware in Django and Python to log metrics and enhance observability across the platform.
- Integrated Prometheus with Django REST Framework (DRF), enabling performance tracking and synthetic monitoring.
- Designed custom Prometheus exporters in Go, facilitating efficient metric collection and visualization.
- Implemented API monitoring and logging solutions, improving system diagnostics and debugging capabilities.

Rolls Royce (AI Hub)

Lead Python & ReactJS Developer
June 2019 – November 2019

- Led development of **internal AI platforms** supporting predictive analytics and decision-making for complex engineering systems.

- Designed and implemented **data ingestion pipelines and APIs** to aggregate data from multiple internal sources for AI and analytics use cases.

- Built Python backend services to support **model execution, results serving, and integration with downstream systems**.

- Developed ReactJS frontends to expose AI outputs and analytical insights to engineering and business users.

- Focused on reliability, data quality, and maintainability in a safety-critical industrial environment.

Monolith AI
Lead Fullstack Developer (Python / TypeScript)
January 2019 – June 2019

Bauer Media Group – UK
Fullstack NodeJS / TypeScript & Python Developer
April 2017 – December 2018

Keytree
Frontend Developer
January 2017 – March 2017

Gameyfi
Lead TypeScript Developer
September 2016 – January 2017

King & Shaxson
Visual Basic & JavaScript Developer
January 2016 – June 2016

Sky
JavaScript Developer
October 2015 – December 2015

Convero
Frontend Developer
March 2015 – September 2015

Chelgate Limited
PHP / JavaScript / CSS Developer
August 2014 – February 2015

Music Career

Musician & Educator
Jazz Double Bassist
September 2004 – January 2014 | Paris, France

- Performed and recorded with notable artists such as Eric Bibb, Pee Wee Ellis, Mel Gaynor, and Paul Jones.
- Jazz educator with extensive teaching experience in London and Paris.

Education

2001- 2004 BA Hons Music (Upper 2:1) - Anglia Ruskin University

1997 -1999 HND in Fine Art - Colchester Institute of Art

References
Available upon request.